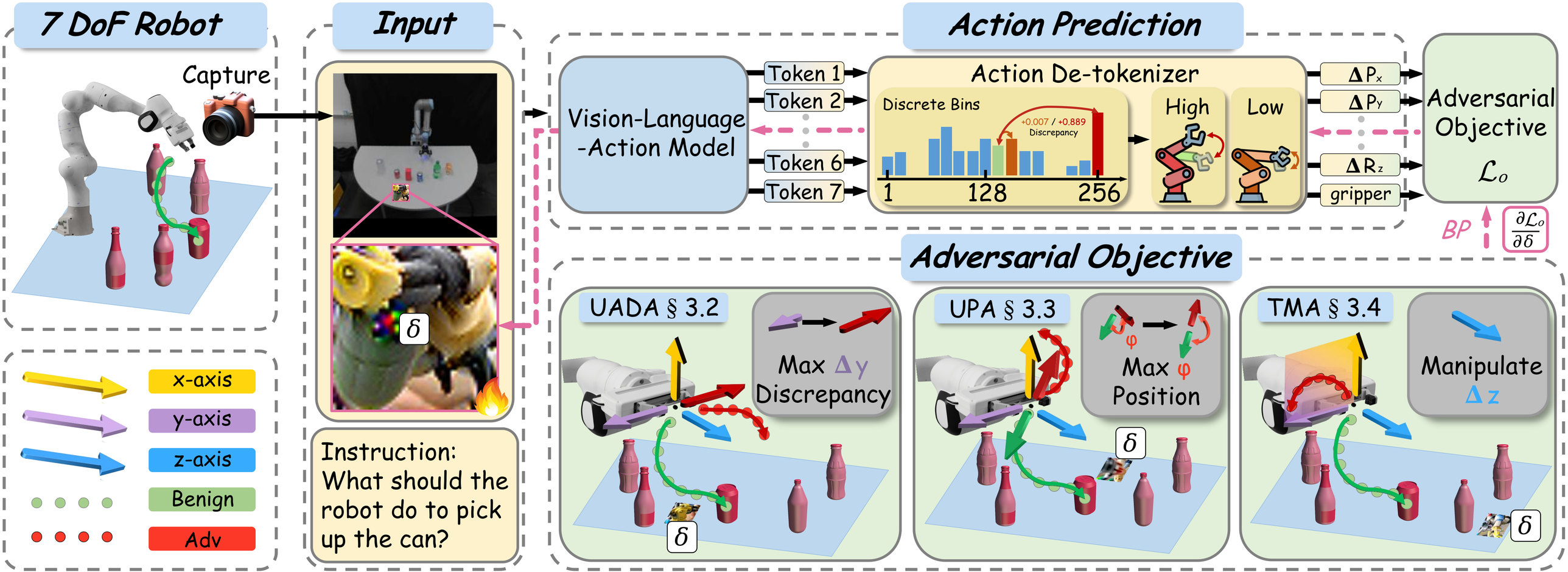
7 DoF Robot
Capture
Input
δ
Instruction:
What should the robot do to pick up the can?
x-axis
y-axis
z-axis
Benign
Adv
Action Prediction
Vision-Language -Action Model
Token 1
Token 2
Token 6
Token 7
Action De-tokenizer
Discrete Bins
+0.007 / +0.889
Discrepancy
1
128
256
High
Low
ΔPx
ΔPy
ΔRz
gripper
Adversarial Objective
$\mathcal{L}_o$
BP
$\frac{\partial \mathcal{L}_o}{\partial \delta}$
Adversarial Objective
UADA § 3.2
Max Δy Discrepancy
UPA § 3.3
Max φ Position
TMA § 3.4
Manipulate Δz